DAMIEN CHEMISTRY OLYMPIAD NOTES

Acid–Base Theories

Conceptual Question Set with Worked Answers

Recommended balance: 80% conceptual : 20% calculation
Coverage: Arrhenius, Brønsted–Lowry and Lewis theories; conjugate pairs; amphiprotic species; non-aqueous acid–base chemistry; strength versus concentration; basicity versus alK$_a$linity; boric acid; solvent levelling.

Foundation for SJChO and advanced H2 Chemistry

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1. Arrhenius Acids and Bases

Concept focus. Acids increase aqueous H_3O^+ ; bases increase aqueous OH^- . Always state whether the definition is being used strictly or operationally.

Question 1 Easy

Classify HCl(aq) and NaOH(aq) under the Arrhenius theory. Write an equation showing how each increases the concentration of the characteristic ion in water.

Answer

HCl is an Arrhenius acid because it increases H_3O^+ in water:

$$HCl + H_2O \rightarrow H_3O^+ + Cl^- .$$

NaOH is an Arrhenius base because it dissociates to increase OH^- :

$$NaOH \rightarrow Na^+ + OH^- .$$

Question 2 Medium

A sample containing 0.0150 mol of $Ca(OH)_2$ is dissolved to make 500 cm^3 of solution. Assuming complete dissociation, calculate $[OH^-]$ and explain why $Ca(OH)_2$ is an Arrhenius base.

Answer

$$Ca(OH)_2 \rightarrow Ca^{2+} + 2\ OH^-$$

Hence $n(OH^-) = 2(0.0150) = 0.0300$ mol. Since $V = 0.500$ dm^3,

$$[OH^-] = \frac{0.0300}{0.500} = 0.0600 mol\ dm^{-3} .$$

It is an Arrhenius base because its dissolution directly increases the hydroxide-ion concentration of water.

Question 3 Hard

Sodium oxide reacts with water according to $Na_2O + H_2O \rightarrow 2\ NaOH$. Is solid Na_2O itself an Arrhenius base? Give a careful answer rather than a one-word classification.

Answer

Under a strict definition requiring the substance itself to dissociate and release OH^- , Na_2O is not a hydroxide and does not directly dissociate into OH^- . Operationally, however, it is commonly classified as a basic oxide and an Arrhenius base because its reaction with water produces NaOH, thereby increasing $[OH^-]$. The example exposes ambiguity in overly rigid formulations of the Arrhenius definition.

2. Limitations of the Arrhenius Model

Concept focus. Test the definition against non-aqueous reactions, bases without hydroxide groups and electron-pair reactions without proton transfer.

Question 4 Easy

Explain why the reaction $NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$ cannot be adequately described using only the Arrhenius theory.

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Answer

The reaction occurs without water, so there is no aqueous production of H_3O^+ or OH^- . It is instead described as Brønsted–Lowry proton transfer: HCl donates H^+ and NH_3 accepts it.

Question 5 Medium

The reaction $BF_3 + NH_3 \rightarrow BF_3 \cdot NH_3$ contains neither transferable protons nor hydroxide ions.
Why is it nevertheless an acid–base reaction, and what limitation of Arrhenius theory does it demonstrate?

Answer

BF_3 accepts a lone pair from NH_3, so BF_3 is a Lewis acid and NH_3 is a Lewis base. Arrhenius theory cannot describe acid–base reactions based on electron-pair transfer or reactions occurring outside aqueous solution.

Question 6 Hard

Critique the statement: "An acid is any substance containing hydrogen, while a base is any substance containing an OH group." Give at least two chemically distinct counterexamples.

Answer

The statement confuses composition with behaviour. CH_4 and C_2H_5OH contain hydro- gen but are not normally acids in water. NH_3 contains no OH group yet is a Brønsted–Lowry and Lewis base. BF_3 contains neither hydrogen nor OH but is a Lewis acid. Conversely, the OH groups in alcohols are covalently bound and do not make alcohols Arrhenius bases.

3. Brønsted–Lowry Proton Transfer

Concept focus. An acid donates H^+ ; a base accepts H^+ . Track one proton from reactant to product.

Question 7 Easy

In $HCl + H_2O \rightarrow H_3O^+ + Cl^-$, identify the Brønsted–Lowry acid and base. State what is transferred.

Answer

HCl is the acid because it donates a proton. H_2O is the base because it accepts that proton. The transferred particle is H^+ ; in solution it is associated with water as H_3O^+ .

Question 8 Medium

For $HCO_3^- + H_2O \rightleftharpoons H_2CO_3 + OH^-$, determine whether water acts as an acid or a base, and justify your choice by tracking the proton.

Answer

Water acts as the acid: it donates H^+ to HCO_3^- , producing OH^- . The bicarbonate ion acts as the base because it accepts H^+ to form H_2CO_3.

Question 9 Hard

Consider $H_2PO_4^- + NH_3 \rightleftharpoons HPO_4^{2-} + NH_4^+$. The relevant values are pK_a $(H_2PO_4^-) = 7.20$ and $pKa(NH_4^+) = 9.25$. Identify the acid and base on each side and estimate the equilibrium constant for the forward reaction.

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Answer

On the left, $H_2PO_4^-$ is the acid and NH_3 is the base. On the right, NH_4^+ is the conjugate acid and HPO_4^{2-} is the conjugate base. For $HA + B^- \rightleftharpoons A^- + HB^+$,

$$K \approx 10^{pKa(HB^+)-pKa(HA)} = 10^{9.25-7.20} \approx 1.1 \times 10^2.$$

Products are favoured because the reaction forms the weaker acid, NH_4^+ , from the stronger acid, $H_2PO_4^-$.

4. Conjugate Acid—Base Pairs

Concept focus. A conjugate pair differs by exactly one H^+ ; charge changes by one unit.

Question 10 Easy

Give the conjugate acid of NH_3 and the conjugate base of H_2O.

Answer

Adding one proton to NH_3 gives its conjugate acid, NH_4^+ . Removing one proton from H_2O gives its conjugate base, OH^- .

Question 11 Medium

In $HSO_4^- + H_2O \rightleftharpoons SO_4^{2-} + H_3O^+$, identify both conjugate acid—base pairs.

Answer

The pairs are HSO_4^-/SO_4^{2-} and H_3O^+/H_2O. Members of each pair differ by exactly one proton.

Question 12 Hard

In water, the hexaaquairon(III) ion undergoes $[Fe(H_2O)_6]^{3+} + H_2O \rightleftharpoons [Fe(H_2O)_5(OH)]^{2+} + H_3O^+$. Identify the conjugate pairs and explain why a metal aquo ion can behave as a Brønsted acid even though the proton originally belongs to a coordinated water molecule.

Answer

The pairs are $[Fe(H_2O)_6]^{3+}/[Fe(H_2O)_5(OH)]^{2+}$ and H_3O^+/H_2O. The highly charged Fe^{3+} ion withdraws electron density from the O—H bonds of coordinated water, increasing their polarity and making proton loss easier. The whole hydrated complex is therefore the proton donor and is treated as the Brønsted acid.

5. Amphiprotic Species

Concept focus. An amphiprotic particle has both a proton it can donate and a site that can accept a proton.

Question 13 Easy

Why is HCO_3^- amphiprotic? Name the species formed when it accepts a proton and when it donates a proton.

Answer

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It can accept H^+ to form H_2CO_3, or donate H^+ to form CO_3^{2-} . Because it can act as both a Brønsted acid and a Brønsted base, it is amphiprotic.

Question 14 Medium

Write two equations with water that demonstrate the amphiprotic behaviour of $H_2PO_4^-$.

Answer

As an acid: $H_2PO_4^- + H_2O \rightleftharpoons HPO_4^{2-} + H_3O^+$. As a base: $H_2PO_4^- + H_2O \rightleftharpoons H_3PO_4 + OH^-$. The first equation shows proton donation; the second shows proton acceptance.

Question 15 Hard

For HS^-, K_a for $HS^- \rightarrow H^+ + S^{2-}$ is approximately 10^{-14}, while K_a for H_2S is approximately 10^{-7}. Use $Kw = 10^{-14}$ to decide whether HS^- shows the stronger tendency to act as an acid or as a base in water.

Answer

Its acidic tendency is measured by $K_a(HS^-) \approx 10^{-14}$. Its basic tendency is

$$K_b(HS^-) = \frac{K_w}{K_a(H_2S)} = \frac{10^{-14}}{10^{-7}} = 10^{-7}.$$

Since $K_b \gg K_a$, HS^- behaves predominantly as a base in water, although it remains amphiprotic.

6. Relative Acid and Base Strength

Concept focus. Proton transfer favours formation of the weaker acid and weaker base; larger K_a corresponds to stronger acid.

Question 16 Easy

Hydrochloric acid is essentially fully ionised in dilute water, whereas ethanoic acid is only partially ionised. Which is the stronger acid, and what does “stronger” mean here?

Answer

HCl is the stronger acid. Acid strength refers to the tendency of an acid particle to donate a proton, represented by the position of its ionisation equilibrium or the magnitude of K_a, not by how much acid is present.

Question 17 Medium

For $CH_3COOH + F^- \rightleftharpoons CH_3COO^- + HF$, use pK_a (CH_3COOH) = 4.76 and pK_a (HF) = 3.17 to estimate K and state which side is favoured.

Answer

$$K \approx 10^{pK_a(HF) - pK_a(CH_3\ COOH)} = 10^{3.17-4.76} \approx 2.6 \times 10^{-2}.$$

The reactant side is favoured. Proton-transfer equilibria favour the side containing the weaker acid and weaker base; here ethanoic acid is weaker than HF.

Question 18 Hard

Arrange $HClO$, $HClO_2$, $HClO_3$ and $HClO_4$ in increasing acid strength. Explain the trend without merely quoting oxidation states.

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Answer

$HClO < HClO_2 < HClO_3 < HClO_4$. Additional oxygen atoms withdraw electron density inductively from the O—H bond and, after deprotonation, allow the negative charge to be delocalised over more oxygen atoms. Both effects stabilise the conjugate base, so proton loss becomes progressively more favourable.

7. Lewis Acids as Electron-Pair Acceptors

Question 19 Easy

In $BF_3 + NH_3 \rightarrow BF_3 \cdot NH_3$, identify the Lewis acid and state why it can accept an electron pair.

Answer

BF_3 is the Lewis acid. Boron has only six electrons around it in the isolated molecule and possesses an empty orbital that can accept the nitrogen lone pair.

Question 20 Medium

Compare the Lewis acidity of $AlCl_3$ and $AlCl_4^-$. Which is the better electron-pair acceptor, and why?

Answer

$AlCl_3$ is the better Lewis acid because aluminium is electron deficient and can accept a lone pair. In $AlCl_4^-$, aluminium has a completed octet and the negative charge also opposes further electron-pair donation, so it is much less Lewis acidic.

Question 21 Hard

A simplistic argument predicts BF_3 to be more Lewis acidic than BCl_3 because fluorine is more electronegative. Yet toward many common donors, BCl_3 is the stronger Lewis acid. Explain this apparent reversal.

Answer

In BF_3, effective $p\pi$-$p\pi$ donation from filled fluorine orbitals into the empty boron p orbital reduces boron's electron deficiency. Back-donation is weaker in BCl_3 because the orbital size and energy match is poorer, leaving boron more able to accept a donor pair. Lewis acidity is therefore controlled by the complete bonding situation and by the chosen donor/solvent, not electronegativity alone.

8. Lewis Bases as Electron-Pair Donors

Question 22 Easy

Which of NH_3, H_2O and Cl^- can act as Lewis bases? Identify the donating electron pair in each case.

Answer

All three can act as Lewis bases. Each donates a non-bonding electron pair: the nitrogen lone pair in NH_3, an oxygen lone pair in H_2O, and a chloride lone pair in Cl^- .

Question 23 Medium

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Predict whether NH_3 or NF_3 is the stronger Lewis base toward BF_3. Explain using electron availability.

Answer

NH_3 is the stronger Lewis base. The highly electronegative fluorine atoms in NF_3 withdraw electron density from nitrogen, making the nitrogen lone pair less available for donation. Hydrogen does not exert such a strong electron-withdrawing effect.

Question 24 Hard

Pyridine and pyrrole both contain a nitrogen lone pair. In pyridine the lone pair lies outside the aromatic π system; in pyrrole it supplies two electrons to complete the six-electron aromatic π system. Which is the stronger Lewis base, and why?

Answer

Pyridine is the stronger Lewis base because its lone pair can be donated without destroying aromaticity. Donation of pyrrole's lone pair would remove two electrons from the aromatic sextet and sacrifice substantial stabilisation, so that pair is much less available.

9. Coordinate-Bond Formation

Concept focus. A donor supplies both electrons to an acceptor; after formation, the bond is not a separate physical category of covalent bond.

Question 25 Easy

Show the formation of NH_4^+ from NH_3 and H^+ using a coordinate-bond arrow. Which atom supplies both bonding electrons?

Answer

$NH_3^+ H^+ \rightarrow NH_4^+$. Nitrogen supplies both electrons in the newly formed N−H bond.

Question 26 Medium

Gaseous $AlCl_3$ can dimerise to Al_2Cl_6. Explain how chloride bridges form using the language of coordinate bonding.

Answer

Each monomeric $AlCl_3$ contains electron-deficient aluminium. A chlorine atom on one unit donates a lone pair into an empty orbital on aluminium of the other unit, producing bridging Al−Cl−Al links. The donor is a chlorine lone pair and the acceptor is aluminium.

Question 27 Hard

In $[Cu(NH_3)_4(H_2O)_2]^{2+}$, identify the Lewis acid, Lewis bases and the number of coordinate bonds. After formation, are these bonds fundamentally different from ordinary covalent bonds?

Answer

Cu^{2+} is the Lewis acid. Four NH_3 molecules and two H_2O molecules are Lewis bases, giving six ligand-to-metal coordinate bonds. The term "coordinate" records that both bonding electrons originated from the ligand; once

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formed, the bond is an ordinary polar covalent interaction and cannot generally be distinguished by examining the electron pair alone.

10. Species Satisfying One Definition but Not Another

Question 28 Easy

Classify BF_3 under the Arrhenius, Brønsted–Lowry and Lewis definitions of an acid.

Answer

BF_3 is a Lewis acid because it accepts an electron pair. It is not itself a Brønsted–Lowry acid because it has no proton to donate, and it is not an Arrhenius acid because it does not dissociate in water to release H_3O^+ .

Question 29 Medium

Classify $NH_3(g)$ as a base under the three theories. Then explain why its classification may appear to change when it is dissolved in water.

Answer

Gas-phase NH_3 is a Brønsted–Lowry base because it accepts H^+ , and a Lewis base because it donates its lone pair. It is not directly an Arrhenius hydroxide. In water, $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$, so it increases $[OH^-]$ and is often treated as an Arrhenius base under the operational, broader formulation.

Question 30 Hard

Classify $CO_2 + OH^- \rightarrow HCO_3^-$ under the three theories. Why is calling CO_2 an "acidic oxide" compatible with its having no proton?

Answer

The carbon atom in CO_2 accepts an electron pair from OH^- , so CO_2 is a Lewis acid and OH^- a Lewis base. The elementary interaction is not Brønsted proton donation by CO_2, and it is not an Arrhenius ionisation. The label "acidic oxide" describes its reaction with bases and formation of acidic species in water; it does not require the oxide molecule itself to contain a transferable proton.

11. Acid–Base Behaviour Outside Aqueous Solution

Question 31 Easy

Explain the acid–base roles in $HCl(g)^+ \; NH_3(g)^- \rightarrow NH_4Cl(s)$ without referring to H_3O^+ or OH^- .

Answer

HCl donates a proton and is the Brønsted–Lowry acid. NH_3 accepts the proton and is the Brønsted–Lowry base. The same event is also Lewis acid–base chemistry because the nitrogen lone pair is donated to H^+ .

Question 32 Medium

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Liquid ammonia undergoes autoionisation: $2\ NH_3 \rightleftharpoons NH_4^+ + NH_2^-$. In this solvent, explain why NH_4Cl behaves as an acid and KNH_2 behaves as a base.

Answer

NH_4Cl supplies NH_4^+ , which increases the concentration of the solvent's characteristic acidic ion and can donate a proton to NH_3. KNH_2 supplies NH_2^- , the characteristic basic ion, which strongly accepts protons. This is the solvent-system analogue of H_3O^+ /OH^- in water.

Question 33 Hard

In glacial ethanoic acid, a very strong acid can protonate the solvent: $HClO_4 + CH_3COOH \rightarrow ClO_4^- + CH_3COOH_2^+$. Identify the acid and base, and explain why an aqueous-only definition is inadequate.

Answer

$HClO_4$ is the proton donor and Brønsted acid; CH_3COOH is the proton acceptor and Brønsted base. The products are ClO_4^- and protonated ethanoic acid. No water is present and no hydroxide ion is involved, so Arrhenius language alone cannot provide a general account.

12. Acid Strength versus Acid Concentration

Question 34 Easy

Two solutions contain 0.100 mol dm^{-3} and 1.00 mol dm^{-3} HCl, respectively. Do they contain acids of different strength? Explain.

Answer

No. Both contain the same acid, so their intrinsic acid strength is the same. The second solution is more concentrated because it contains more HCl per unit volume. Strength describes extent or tendency of ionisation; concentration describes amount per volume.

Question 35 Medium

A 1.0 mol dm^{-3} solution of ethanoic acid and a 0.010 mol dm^{-3} solution of hydrochloric acid are compared. State separately which acid is stronger and which solution contains more acid particles per unit volume.

Answer

Hydrochloric acid is intrinsically stronger because its proton donation in water is essentially complete. The ethanoic-acid solution is more concentrated and contains more acid formula units per unit volume. A concentrated weak acid is not thereby converted into a strong acid.

Question 36 Hard

A weak monoprotic acid HA has $K_a = 1.0 \times 10^{-5}$ and concentration 0.100 mol dm^{-3}.
Compare its equilibrium $[H_3O^+]$ with that of a $5.0 \times 10^{-}4$ mol dm^{-3} strong monoprotic acid. Use the weak-acid approximation but do not calculate pH.

Answer

For the weak acid, $[H_3O^+] \approx \sqrt{(K_a c)} = \sqrt{(1.0 \times 10^{-5})(0.100)} = 1.0 \times 10^{-3}$ mol dm^{-3}.

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The dilute strong acid gives approximately 5.0×10^{-4} mol dm^{-3}. Thus the weaker acid solution has the larger hydronium concentration because it is far more concentrated. This does not make HA the stronger acid.

13. Basicity versus Alkalinity

Question 37 Easy

Explain why CuO is a base but not an alkali, whereas NaOH is both.

Answer

CuO neutralises acids, so it is a base, but it is insoluble in water and does not produce an alkaline aqueous solution. NaOH is a soluble base; it dissolves to form OH^- , so it is an alkali. An alkali is therefore a water-soluble base.

Question 38 Medium

One litre of 0.0100 mol dm^{-3} NaOH and one litre of 0.200 mol dm^{-3} NH_3 are separately neutralised by a strong acid. Which contains the intrinsically stronger base particle, and which can neutralise more moles of H^+ ?

Answer

The stronger base particle is OH^- ; it has a greater tendency to accept H^+ than NH_3. However, the NaOH solution contains only 0.0100 mol of OH^- , whereas the ammonia solution contains 0.200 mol of NH_3. Assuming complete stoichiometric protonation, the ammonia solution can neutralise more H^+ . Intrinsic basicity and total acid-neutralising capacity are different ideas.

Question 39 Hard

A 250 cm3 solution contains 0.0200 mol dm^{-3} CO_3^{2-} and 0.0300 mol dm^{-3} HCO_3^- . Calculate the maximum moles of H^+ it can consume when both species are converted fully to H_2CO_3. Relate the result to alkalinity.

Answer

n(CO_3^{2-}) = 0.0200(0.250) = 0.00500 mol, consuming 2(0.00500) = 0.0100 mol of H^+ .
n(HCO_3^-) = 0.0300(0.250) = 0.00750 mol, consuming 0.00750 mol of H^+ .
Total capacity: 0.0100+ 0.00750 = 0.0175 mol H^+.
This equivalent acid-neutralising capacity is the solution's alkalinity for the stated endpoint; it depends on amounts and stoichiometry, not only on the intrinsic strength of each base.

14. Olympiad Extension: Boric Acid as a "Weird" Acid

Question 40 Easy

Boric acid reacts with water as follows: $B(OH)_3 + 2\ H_2O \rightleftharpoons B(OH)_4^- + H_3O^+$. Explain why $B(OH)_3$ is acting primarily as a Lewis acid rather than by directly donating one of its own O—H protons.

Answer

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Electron-deficient boron accepts an electron pair from a water-derived hydroxide donor, forming tetrahedral $B(OH)_4^-$. The associated proton transfer between water molecules produces H_3O^+ . Thus acidity arises from acceptance of OH^- /an oxygen lone pair at boron, not simple loss of a proton directly bonded to boric acid.

Question 41 Medium

A sample contains 0.0100 mol of $B(OH)_3$. How many moles of NaOH are required for complete neutralisation according to its aqueous mechanism? Explain why the answer is not three times larger.

Answer

The stoichiometry is one mole of OH^- per mole of $B(OH)_3$: $B(OH)_3 + OH^- \rightarrow B(OH)_4^-$. Therefore 0.0100 mol of NaOH is required. The three written OH groups are not three indepen- dently ionisable acidic protons; boric acid behaves as a monobasic Lewis acid in water.

Question 42 Hard

Boric acid is placed in water enriched with $H_2{}^{18}O$. Using the Lewis-acid mechanism, predict whether an ^{18}O label can become incorporated into $B(OH)_4^-$. Explain what this mechanistic test distinguishes from a simple direct-proton-donation picture.

Answer

Yes. Because a water oxygen donates to boron and becomes part of the tetrahedral borate environment, oxygen exchange or direct incorporation of solvent-derived ^{18}O into $B(OH)_4^-$ is mechanistically plausible. This supports addition of a solvent-derived hydroxyl group at boron. A simplistic model in which $B(OH)_3$ merely loses one of its own protons would not, by that elementary step alone, require incorporation of a new solvent oxygen into the conjugate base.

15. Olympiad Extension: Solvent Levelling and Differentiation

Question 43 Easy

Why do HCl, HBr, HI and $HClO_4$ all appear essentially completely ionised in dilute aqueous solution, even though their intrinsic gas-phase acidities are not identical?

Answer

Water is sufficiently basic to accept protons essentially completely from all of them. Each is converted to the strongest acid that can persist in appreciable concentration in water, H_3O^+ . Their strengths are therefore levelled to that of H_3O^+ in this solvent.

Question 44 Medium

Liquid ammonia autoionises to NH_4^+ and NH_2^- . Identify the strongest acid and strongest base that can persist in this solvent, and explain what happens when a much stronger acid or base is added.

Answer

The levelling acid is NH_4^+ ; any stronger acid protonates NH_3 and is converted to NH_4^+ . The levelling base is NH_2^- ; any stronger base removes a proton from NH_3 and is converted to NH_2^- . Thus the solvent's own conjugate acid/base pair sets the observable acid−base window.

Question 45 Hard

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Water cannot readily distinguish the strengths of $HClO_4$ and HNO_3, but a less basic solvent such as glacial ethanoic acid may do so. Explain what makes a solvent "differentiating" and why changing solvent can reverse or reveal apparent strength orderings.

Answer

A differentiating solvent does not protonate completely in response to every acid being compared. Proton-transfer equilibria therefore remain at measurably different positions, revealing differences in acidity. Solvation of the acid, conjugate base and protonated solvent also changes from one medium to another, so observed strength is not an immutable property of the isolated formula alone. A solvent can level acids stronger than its conjugate acid while differentiating weaker ones.

End of question set

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